

Short-term Traffic Flow Combined Forecasting Model Based on SVM

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Abstract—This research concerns itself with a wavelet-SVM combined model study of the short-term traffic flow prediction issue. Different theories and methods have been introduced in the field to solve short-term traffic flow forecasting problem. And in our study, we attempt to use an alternative prediction framework to examine the combined model. This paper consists of four sections. A brief introduction is given in Section one of this study. Section two includes the theories of wavelet and support vector machine (SVM), then put forward the combined model. Section three focuses on a numerical study based on the actual speed data of an expressway in Beijing. The whole paper ends with the conclusion that the combined model has very high accuracy.

Keywords- Support vector machine (SVM); Short-term traffic flow; Forecasting; Wavelet; Combined model

I. INTRODUCTION

Intelligent Transportation Systems (ITS) is a time-dependent and complicated system which has characteristic of remarkable uncertainty and non-linearity so as to result in difficulties for real-time traffic flow forecasting, especially for short-term traffic flow forecasting (Wang and Fang, 2009). During the past few years, various short-term traffic flow forecasting models have been developed and forecasting accuracy has been improved greatly. These models include state space models (Chen et al, 1994), time series analysis and neural network models (Han and Song, 2003). However, the method of neural network has some deficiencies such as slow convergence, structure selection problem and so on. To make up for these deficiencies of neural network method, Vapnik introduces support vector machine (SVM) based on statistical learning theory (Vapnik, 1995), which has good generalization ability in solving some practical problems including local optimization, over-fitting and high dimension.

This paper combined wavelet analysis method with support vector machine (SVM), then proposed a short-term traffic flow forecasting combined model based on SVM. Furthermore, we used the detected velocity data of Beijing expressway to validate the combined model. By comparison

analysis of the prediction results of the combined model and traditional SVM model, we got some valuable conclusions.

II. WAVELET-SVM COMBINED MODELS

A. Wavelet Analysis method

With the interference from external factors, short-term traffic flow composition is complicated, which can be considered as the synthesis of multiple signals. Different signal components have different properties and have different influence on the traffic flow changing trend (Yu, 2005). Deterministic inherent signals decide the traffic flow changing trend in the overall. Uncertain interference signals bring about the different levels of fluctuation near the actual traffic flow changing trend. Traffic flow uncertain interference signal frequency components are more complex and change more rapidly. Various interference signals mixing together increase greatly the difficulty of short-term traffic flow forecasting (Yang, F.2S, 2000). In order to improve the accuracy of prediction, the first thing is signal analysis, transforming the complex short-term traffic flow data sequence into several simple data sequences. In this paper, we use "wavelet" to decompose traffic flow signal and extract traffic flow characteristic signals under different frequency.

Short-term traffic flow signals are equivalent to a signal set V_0 composed of several original signals. Conducting wavelet decomposition and reconstruction on the data sequence V_i ($V_i \in V_0$), we can obtain $N+1$ corresponding signals. W_i^j is the interference signal between V_i^j and V_i^{j+1} . In addition, $W_i^1, W_i^2, \dots, W_i^N, V_i^N$ are independent of each other (Qu et al. 2007). These $N+1$ signals constitute the original signal $V_i = W_i^1 \oplus W_i^2 \oplus \dots \oplus W_i^N \oplus V_i^N$. According to multi-resolution analysis principle, we can decompose and reconstruct a short-time traffic flow signal to a basic signal V_i^N and N independent interference signals W_i^N . V_i^N expresses the fundamental changing trend of short-time traffic flow.

Mallet put forward tower multi-resolution signal decomposition and reconstruction algorithm, the basic idea is if the discrete approximation of a signal $f(t)$ is $A_j f(t)$ under resolution 2^{-j} , then the discrete approximation $A_{j+1} f(t)$ of $f(t)$ can be obtained by discrete low-pass filter (Yu, 2005).

$\varphi_{-j,k}(t)$ and $\psi_{-j,k}(t)$ are respectively the scaling function and wavelet function of $f(t)$ under resolution 2^{-j} then the discrete approximation $A_j f(t)$ and detail $D_j f(t)$ can be expressed as

$$A_j f(t) = \sum_{k=-\infty}^{\infty} C_{j,k} \varphi_{-j,k}(t), \quad D_j f(t) = \sum_{k=-\infty}^{\infty} D_{j,k} \psi_{-j,k}(t), \quad j > 0 \quad (1)$$

Where $C_{j,k}$ and $D_{j,k}$ are the approximate component decomposition coefficients and detail component decomposition coefficients. According to Mallat decomposition algorithm, $A_j f(t) = A_{j+1} f(t) + D_{j+1} f(t)$. Through the decomposition and synthesis iteration of $C_{j,k}$ and $D_{j,k}$. The Mallat pyramid decomposition algorithm and the reconstruction algorithm are achieved (Yu, 2005).

B. SVM prediction theory

Support Vector Machine (SVM) is an advanced machine learning method, which is based on statistical learning theory and structural risk minimization principle. By means of data mining, SVM can find out the regular and objective patterns behind of data. These patterns can not be obtained by current theory analysis. SVM can predict the unknown data according to those patterns (Wang and Fang, 2009). Furthermore, SVM has a good learning ability and generalization ability superior to neural network. It is easy for SVM to achieve balance between fitting degree and generalization level (Yang and Liu, 2007). Due to these advantages, SVM has been applied to various research fields. Especially SVM regression (SVR) application studies show good performance of the intelligent algorithm. Through a nonlinear transformation, SVM maps the input variables into a higher dimensional feature space and construct a separating optimal hyper plane with maximum margin (Wang et.al. 2008).

Assuming a training data sample $(x_1, y_1), \dots, (x_l, y_l) \in (X \times R)$, is generated according to some probability distribution $P(x, y) (x \in R^n, y \in R)$, where l denotes the size of the training data. Support vector regression is to determine a suitable function which can fit the training data by minimizing expected risk $R[f]$,

$$R[f] = \int c(x, y, f) dP(x, y), \quad (2)$$

Where c represents cost function.

The SVR function is formulated as $f(x) = w \cdot \phi(x) + b$, where $w \in R^n$, $b \in R$, $\phi(x_i)$ is a nonlinear transformation from R^n to high-dimensional space.

We use insensitive function to measure the error between the true value and the predicted value (C. H, 2004), that is

$$|y_i - f(x_i, x)|_{\varepsilon} = \max\{0, |y_i - f(x_i)| - \varepsilon\}, \quad (3)$$

However, we are unable to figure out the minimum of $R[f]$ since $P(x, y)$ is unknown. This problem is solved by minimizing $E(w)$,

$$E(w) = \frac{1}{2} (w \cdot w) + \frac{C}{l} \sum_{i=1}^l |y_i - f(x_i)|_{\varepsilon}, \quad (4)$$

Where $(w \cdot w)$ describes the complexity of the function $f(x)$. C reflects a compromise relationship between the complexity of the function $f(x)$ and the average cost of training set (Yang and Liu, 2007).

With Slack variables ξ_i and ξ_i^* being introduced, equation (4) is converted into an equivalent programming model.

$$\begin{aligned} \min_{w, \xi_i, \xi_i^*, b} \quad & \frac{1}{2} (w \cdot w) + \frac{C}{l} \sum_{i=1}^l (\xi_i + \xi_i^*) \\ \text{s.t.} \quad & (w \cdot \phi(x_i) + b) - y_i \leq \varepsilon + \xi_i, \\ & y_i - (w \cdot \phi(x_i) + b) \leq \varepsilon + \xi_i^* \end{aligned} \quad (5)$$

The above model is solved by solving the Lagrange dual problem of the original problem, the dual form of the above model is followed as

$$\begin{aligned} \min_{\alpha, \alpha^*} \quad & \frac{1}{2} \sum_{i=1}^l \sum_{j=1}^l (\alpha_i - \alpha_i^*)(\alpha_j - \alpha_j^*) K(x_i, x_j) \\ & - \sum_{i=1}^l [\alpha_i^*(y_i - \varepsilon) - \alpha_i(y_i + \varepsilon)] \\ \text{s.t.} \quad & \sum_{i=1}^l (\alpha_i - \alpha_i^*) = 0, \quad 0 \leq \alpha_i, \alpha_i^* \leq C/l, \quad i = 1, \dots, l \end{aligned} \quad (6)$$

Where $K(x_i, x_j)$ is the kernel function satisfying the Mercer condition, which includes linear kernel function, polynomial kernel function and Gaussian kernel function. We use Gaussian kernel function as the core function in the research.

Solving the above quadratic programming problem, we can obtain (α, α^*) , then

$$f(x) = w \cdot \phi(x) + b = \sum_{i=1}^l (\alpha_i - \alpha_i^*) K(x_i, x) + \bar{b}, \quad (7)$$

$$\text{Where } b_i = y_i - \sum_{j=1}^l (\alpha_j - \alpha_j^*) K(x_j, x_i) \pm \varepsilon, \quad \bar{b} = \frac{1}{l} \sum_{i=1}^l b_i,$$

C. Combined model of short-term traffic flow forecasting and evaluation index

In this paper, we firstly make use of wavelet analysis de-noise function to remove the noise in traffic flow data, getting a new time-series data. Then the data is multi-level decomposed by wavelet decomposition function and wavelet decomposition coefficients are reconstructed to get the approximate sequence and details sequences, which are predicted by SVM respectively (Zhuo,2008; Qu,2007). Finally, we synthesize the prediction results of each reconstructed sequence.

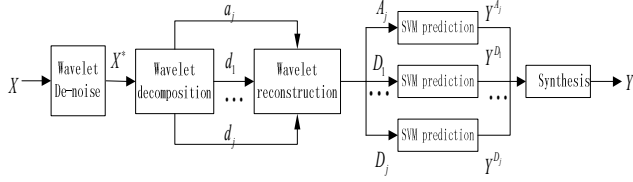


Figure 2.1 the wavelet-SVM combined forecasting model

... We use Relative Mean Errors and Root-Mean-Squared Errors as the Evaluating indexes for the final prediction result.

$$RME = \frac{1}{n} \sum_{i=1}^n \left| \frac{Y_i - Y_i^*}{Y_i^*} \right| \quad (8)$$

$$RMSE = \sqrt{\frac{1}{n} \sum_{i=1}^n \left(\frac{Y_i - Y_i^*}{Y_i^*} \right)^2} \quad (9)$$

Where Y_i denotes the forecasting short-term traffic flow. Y_i^* denotes the actual traffic flow.

III. NUMERICAL EXPERIMENTS

We use the detected velocity data of Beijing expressway to test the proposed combined model. The original data is shown in figure 3.1. the numerical experiment processing is (1)wavelet de-noise; (2) data normalization;(3) wavelet decomposition and reconstruction with signal'sym8', getting $A4, D1, D2, D3, D4$; (4) SVM training data fitting and forecasting of $A4, D1, D2, D3, D4$; (5) synthesize the prediction results of each reconstructed sequence and computer the forecasting error with evaluating indexes above. (6) comparison analysis the prediction results of the combined model and traditional SVM model.

(1)Wavelet de-noise

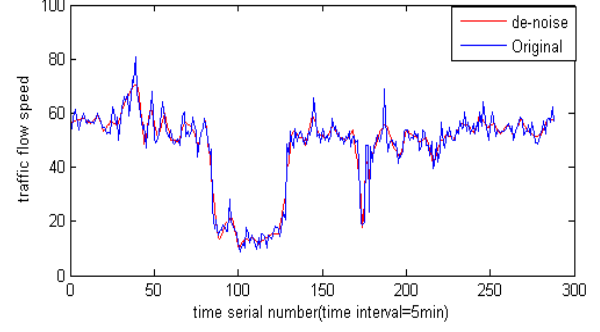


Figure 3.1 The comparison between the de-noise speed data and the original data

(2) Data normalization

Normalized can improve the convergence rate, shorten the training time. Normalization method in this paper is formulated as follow:

$$x_i = \frac{X_i - X_{\min}}{X_{\max} - X_{\min}} \quad (10)$$

(3)Wavelet decomposition and reconstruction with signal'sym8'

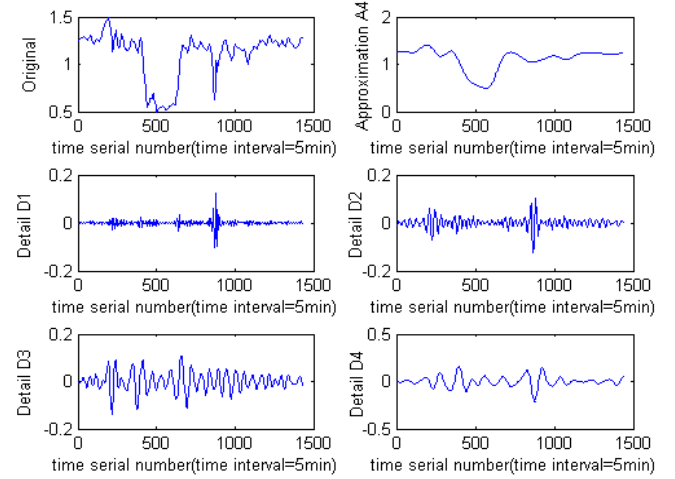


Figure 3.2 The wavelet decomposition and reconstruction

(4) SVM training data fitting and forecasting of $A4, D1, D2, D3, D4$

The length of the data is 288. We use the data from 1 to 144 as training set, and use the data from 145 to 216 as testing set. The fitting and forecasting results of $A4$, is shown in figure 3.4.

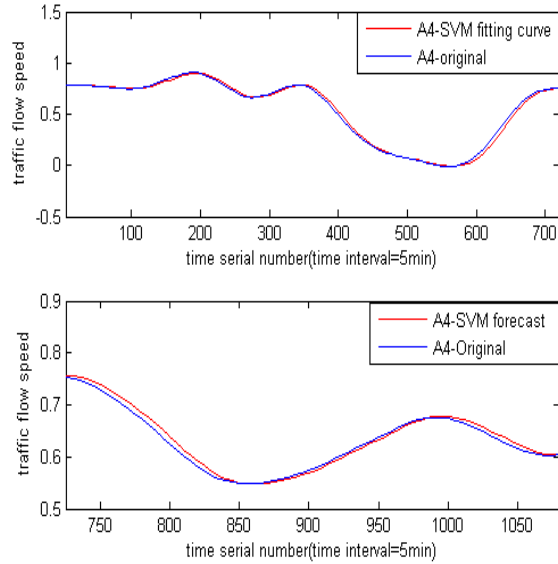


Figure 3.3 A4-SVM fitting and forecasting curve

(5) Synthesize the prediction results of each reconstructed sequence and compute the forecasting error as follows

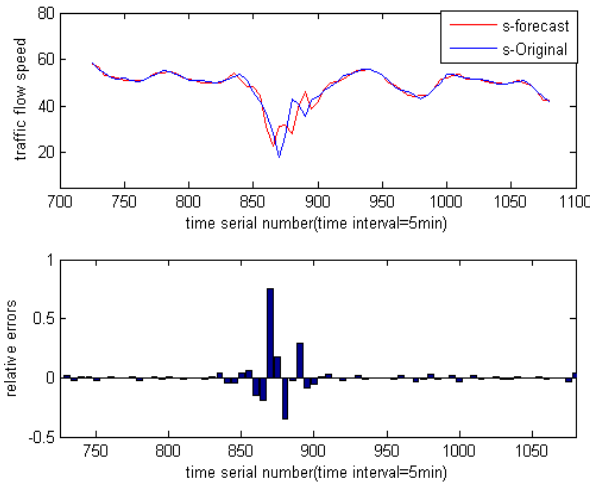


Figure 3.4 the final synthesis forecasting result and error curve

Table 1 the combined model and SVM prediction accuracy contrast

Name	Wavelet-SVM combined model	SVM
RME	4.12%	5.68%
RMSE	11.17%	10.95%

With the prediction errors comparison of the combined model and single SVM method, we can find out when the traffic flow speed is relatively high and changes slowly, the prediction accuracy of the proposed combined model is much higher than single SVM. However, when the traffic flow speed is relatively low and changes dramatically, the prediction accuracy of the proposed combined model is lower than single SVM. This phenomenon is probably caused by the D1

sequence changing too dramatically. The fitting and forecasting errors of D1 are both higher. All in all, the overall predicting effect is perfect, and this combined model has effectively improved the mean forecasting accuracy.

IV. CONCLUSIONS

In this paper, we studied and discussed the short-term traffic flow forecasting issue, which is the traffic flow theory basis for ITS. In terms of SVM's advantage on short-term traffic flow prediction and wavelet analysis for traffic flow signal processing effectively, this paper proposed a wavelet-SVM combined forecasting model. We used the actual traffic flow speed data of an expressway to verify this combined model. The prediction results and the comparison results showed that this combined model for short-term traffic flow prediction is feasible and effective. In addition, the combined model has a problem that the prediction accuracy is a little low if the speed is low and fluctuating. We will start further research on this problem.

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